UATMMS
Traffic Intelligence

ADMIN CONSOLE

- Overview
- Users & Roles
- Safety Policies
- Analytics

Rahul Deshmukh
System Admin · Sign out

Analytics & Reports

Baseline: fixed-time signals · Treatment: UATMMS (demo data until pilot)

Search junction, vehicle...

System Online · AI Operational

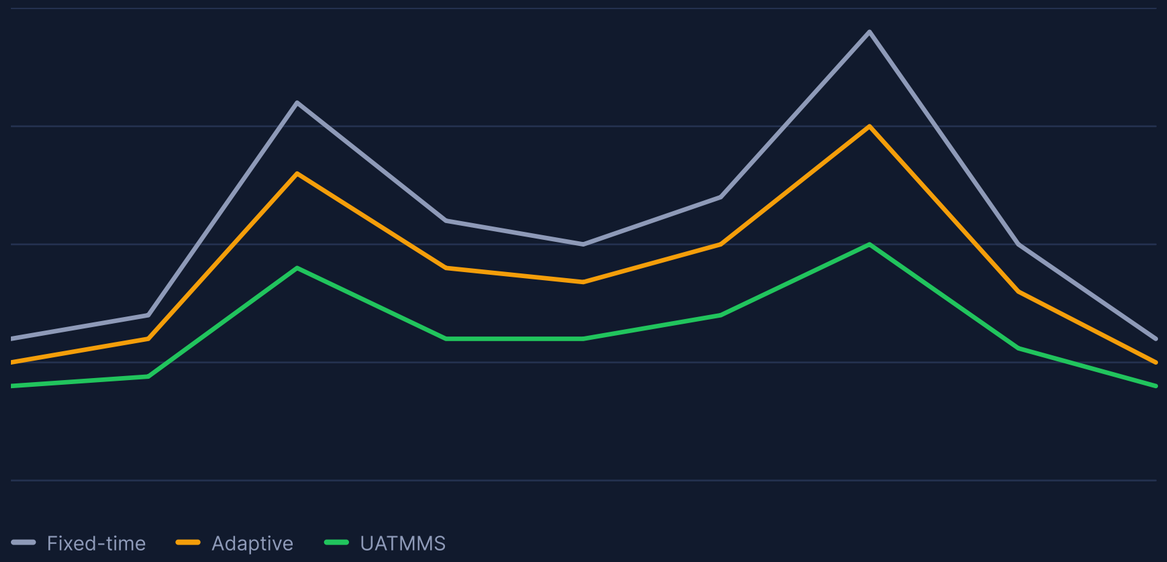
10:24 AM · Thu, 15 Oct 2026

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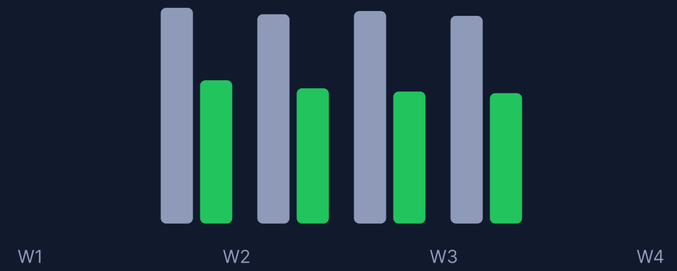
Avg. waiting time	Avg. travel time	Throughput	Emergency travel time	Safety violations
↓ 42%	↓ 35%	↑ 18%	↓ 38%	0
28 s vs 48 s	6.3 vs 9.7 min	3,420 veh/hr	8.1 vs 13.1 min	30-day total

Avg. waiting time over 24 h

Fixed-time · Adaptive · UATMMS



Emergency response time · weekly (min)



Export report

PDF summary	Download ↓
CSV raw data	Download ↓
Pilot comparison	Download ↓

Administration Overview

System health · devices · AI models · audit

System uptime (30 d)
99.98%
Target ≥ 99.9%

Junctions under control
24 / 24
0 in fallback mode

Cameras online
126 / 128
2 need maintenance

Emergency corridors today
17
Avg. 3.4 min saved (demo)

Subsystem Health

All checks 10 s ago

Perception · YOLO + DeepSORT 84 ms / frame	Healthy
Prediction · XGBoost / LSTM MAE 3.1 veh	Healthy
Network optimizer 1.2 s / cycle	Healthy
Safety & fallback engine 0 violations	Healthy
Signal controller gateway 24 / 24 linked	Healthy
Emergency GPS ingest 3.1 s delay	Degraded
Database & cloud Replica lag 120 ms	Healthy

Recent Audit Log

View all

10:21	S. Patil verified incident at J12
10:14	Corridor #A1 created (8 min ETA)
10:02	P. Kulkarni approved J7 timing +10 s
09:47	Model v2.3 prediction retrained
09:30	Safety policy: min green 10 s unchanged
09:12	New user invited: Sneha More

Device Fleet

CCTV cameras	126 / 128
Road sensors	212 / 220
Signal controllers	24 / 24
Emergency GPS units	38 / 40

+ Register device

Safety Constraints & Fallback Policy

Hard rules the AI can never override

Hard Safety Constraints

Enforced by Safety Engine

Minimum green time	10 seconds	Pedestrian clearance time	12 seconds
Maximum cycle length	120 seconds	All-red clearance	2 seconds
Max corridor pre-emption per junction	45 seconds	Max cross-traffic delay increase	15 %

Emergency Priority Rules

When multiple emergency vehicles overlap, priority is resolved in this order:

1. Ambulance · critical patient
2. Fire truck · active fire
3. Police escort · VIP / special
4. Other authorised vehicles

Fallback & Failure Handling

- Auto-fallback to fixed-time on sensor loss — on
- Fallback when model confidence < 70% — on
- Fallback after 3 failed safety validations — on
- Require operator approval for corridors > 5 junctions — off
- Require 2-person approval for manual overrides — off

Policy Status

Policy version	v1.4 · active
Last changed	Rahul Deshmukh · 3 days ago
Violations (30 d)	0
Fallback activations (30 d)	2 (sensor outage)

Save & publish v1.5

Discard

Users & Role Permissions

Role-based access control (RBAC) · invite · suspend

Users

7 users

Search name, role, zone...

+ Invite user

Name	Role	Zone / Org	Status	Last active
RD Rahul Deshmukh	System Admin	HQ	Active	Just now
PK Priya Kulkarni	Traffic Operator	Zone A	Active	2 min ago
IS Insp. S. Patil	Traffic Police	Zone A	Active	8 min ago
DA Dr. A. Joshi	Hospital ER	City Hospital	Active	15 min ago
A# Ambulance #A1 driver	Ambulance Driver	Fleet 108	On duty	Now
Fc Fire crew #F1	Firefighter	Station 2	On duty	Now
SM Sneha More	Traffic Operator	Zone B	Invited	—

Role Permissions

Edit

Permission	Admin	Operator	Police	Emergency	Hospital	Public
View live map	✓	✓	✓	✓	✓	✓
Control signals	✓	✓	–	–	–	–
Raise emergency	✓	–	–	✓	✓	–
Verify incidents	✓	✓	✓	–	–	–
Manage users	✓	–	–	–	–	–
Edit policies	✓	–	–	–	–	–

Public users can only view traffic and receive alerts. Emergency roles can raise priority requests but cannot edit signal timing — only the safety engine can.

Emergency Coordination

City Hospital · ER dispatch & arrival readiness

Incoming Ambulances

1 active

Ambulance #A1

CRITICAL

Male, ~58 · chest pain · triage: critical

ETA	8 min 12 s (live)
Position	Between J7 and J3
Green corridor	Active · 3 / 6 cleared
Assigned bay	ER-2 · cardiac team

Request Ambulance + Green Corridor

Pickup location

Shivaji Chowk, Kolhapur — Pin

Destination

City Hospital · ER

Severity

Critical | High | Normal

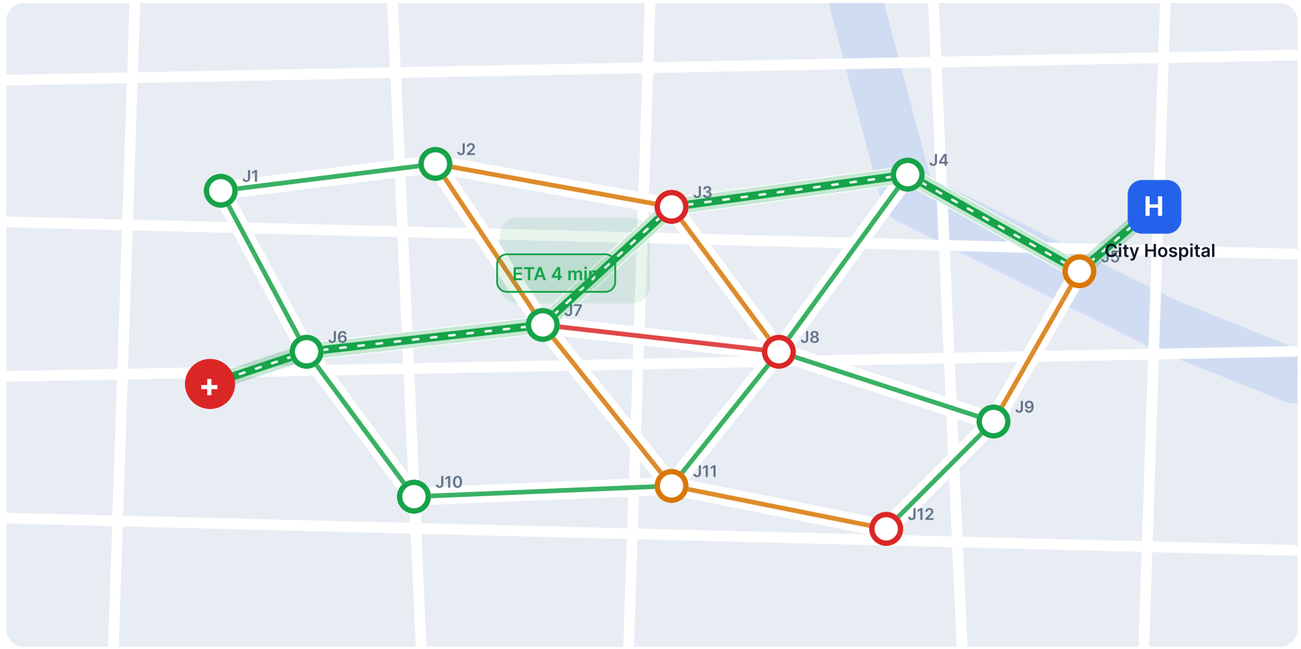
Patient notes (optional)

Chest pain, conscious

Dispatch + create green corridor

Live ambulance tracking

Updated 1 s ago



Bed Availability

Live

Emergency beds	4 / 12 free
ICU	2 / 8 free
Operation theatre	1 / 3 free
Trauma bays	2 / 4 free

ER Readiness for #A1

- [x] Cardiac team alerted
- [x] Bay ER-2 cleared
- [ ] Blood bank notified
- [ ] Triage nurse at entrance

Notify ER team

Emergency Green Corridor

Ambulance #A1 → City Hospital · corridor ACTIVE

Active Emergencies

2 active

Ambulance #A1 — CRITICAL

Shivaji Chowk → City Hospital

48 km/h · 3 of 6 junctions cleared — ETA 8:12

Fire Truck #F1 — HIGH

Fire Station 2 → Rajarampuri Fire Incident

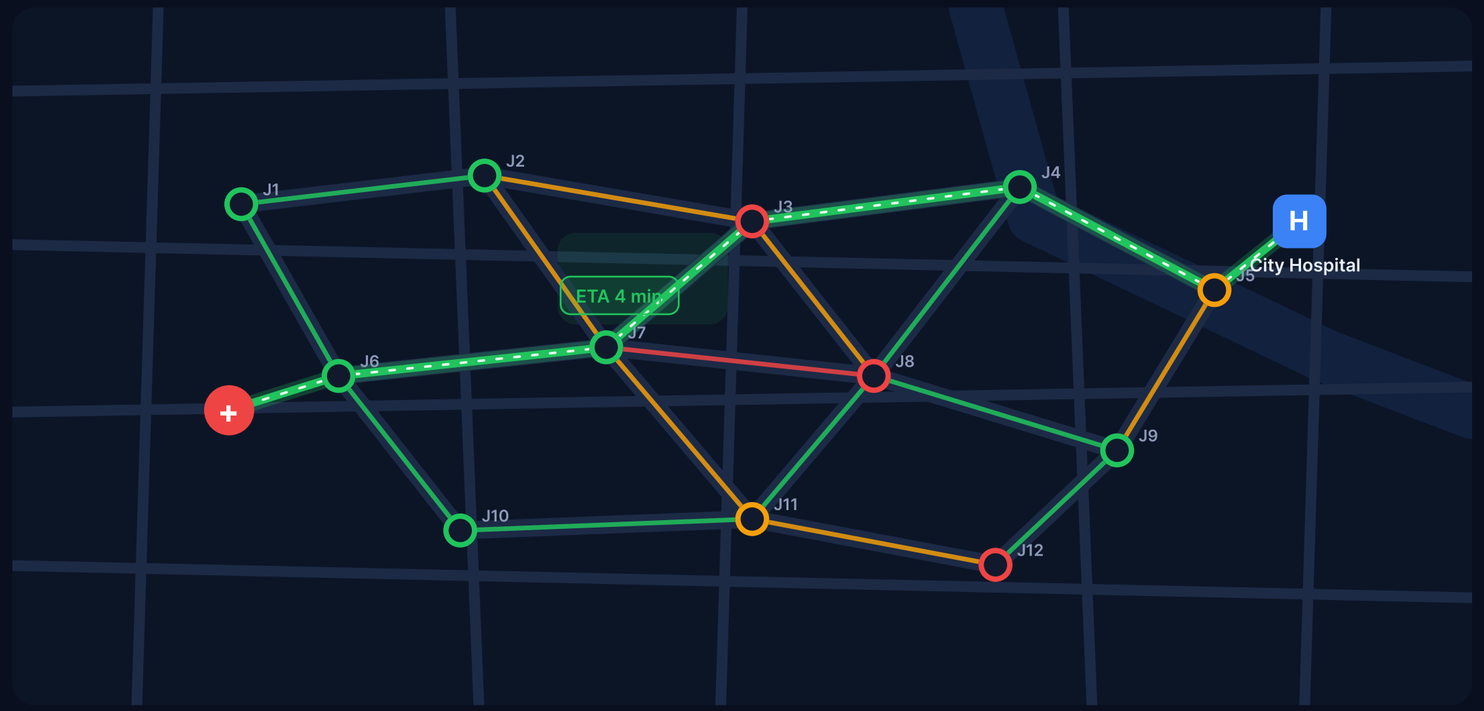
Queued behind #A1 on shared segment J7–J3 · ETA 7 min

+ New emergency request

Corridor Details

Route length	6.2 km
Predicted ETA	8 min 12 s
Time saved vs. normal	≈ 3 min 40 s (demo)
Signals pre-cleared	3 / 6
Cross-traffic impact	Low (bounded)
Fallback timing	Ready

Objective: minimize emergency travel time + bound disruption to surrounding traffic.



Junction Sequence · ETA & signal state

Updated 1 s ago

J6	J7	J3	J4	J5	H
passed	ETA 1:20	ETA 3:05	ETA 5:10	ETA 6:40	ETA 8:12
Cleared	Green held	Pre-clearing	Scheduled	Scheduled	City Hospital

Operator Controls

Hold green +10 s | Re-route | Pause corridor | Abort & restore normal

✓ No conflicting phases · ✓ Pedestrian clearance kept · ✓ Min green respected · ✓ Queue spillback < limit

Incident Detection & Response

Computer-vision + traffic-anomaly detection · human verification

Incident Feed

4 total

All | Open | Verified | Resolved

Accident — 2 min ago
J12 · Vision AI 94%
Needs verification

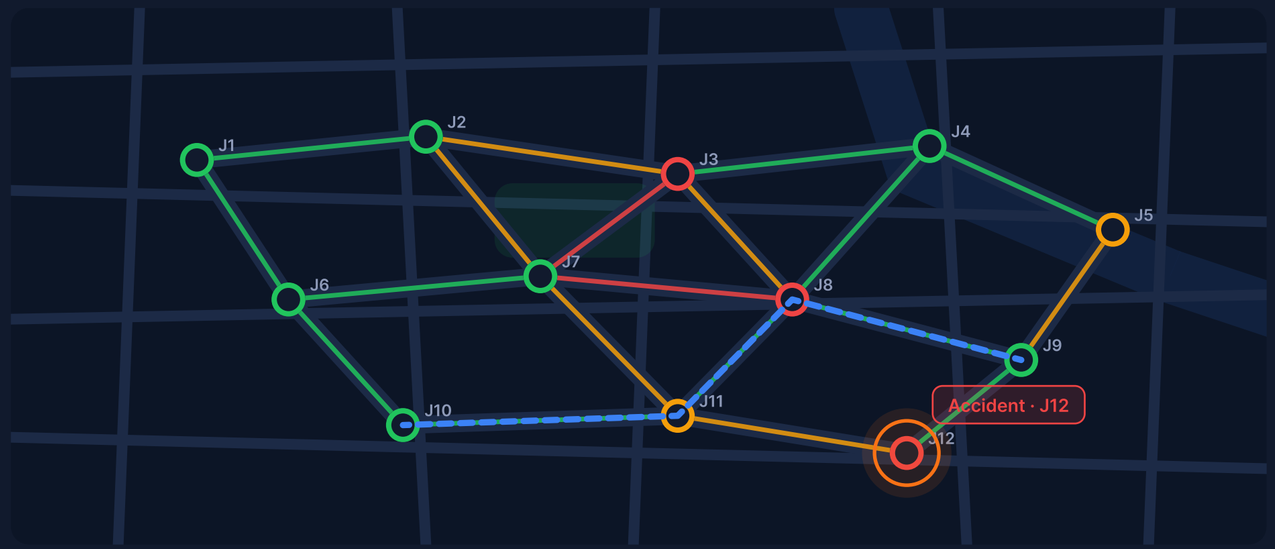
Stalled vehicle — 11 min ago
J8 · Vision AI 81%
Verified

Abnormal slowdown — 18 min ago
J3 · Anomaly model 76%
Auto-diverted

Waterlogging / blockage — 1 hr ago
J10 · Police report
Resolved

Incident location · Junction J12

Diversion proposed



AI Verification

Vision + rules

94% Incident confidence
ML model + rule-based check

- 3 vehicles stopped suddenly in lane 2
- Lane blockage persisted for 60 s
- Rule check passed: not a signal queue

Confirm incident | False alarm

Response Plan

1. **Identify affected area**
   J12 · J11–J12 segment
2. **Recalculate flow & alternate routes**
   Via J9 → J8 → J11
3. **Adjust signal timings**
   Awaiting approval
4. **Notify authorities**
   Traffic Police control room

Notify Traffic Police

UATMMS
Traffic Intelligence

OPERATOR CONSOLE

- Live Map
- Emergency
- Incidents
- Signal Control

Live Traffic Control Center

Kolhapur City Network · 24 intersections · 128 cameras

Search junction, vehicle...

System Online · AI Operational

10:24 AM · Thu, 15 Oct 2026

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Network Overview

Intersections	24
Cameras online	126 / 128
Vehicles tracked	1,842
Network load	72%

Active Emergency Vehicles 2

- A — Ambulance #A1 — To City Hospital · ETA 4 min — Priority
- F — Fire Truck #F1 — To Fire Incident · ETA 7 min — Priority

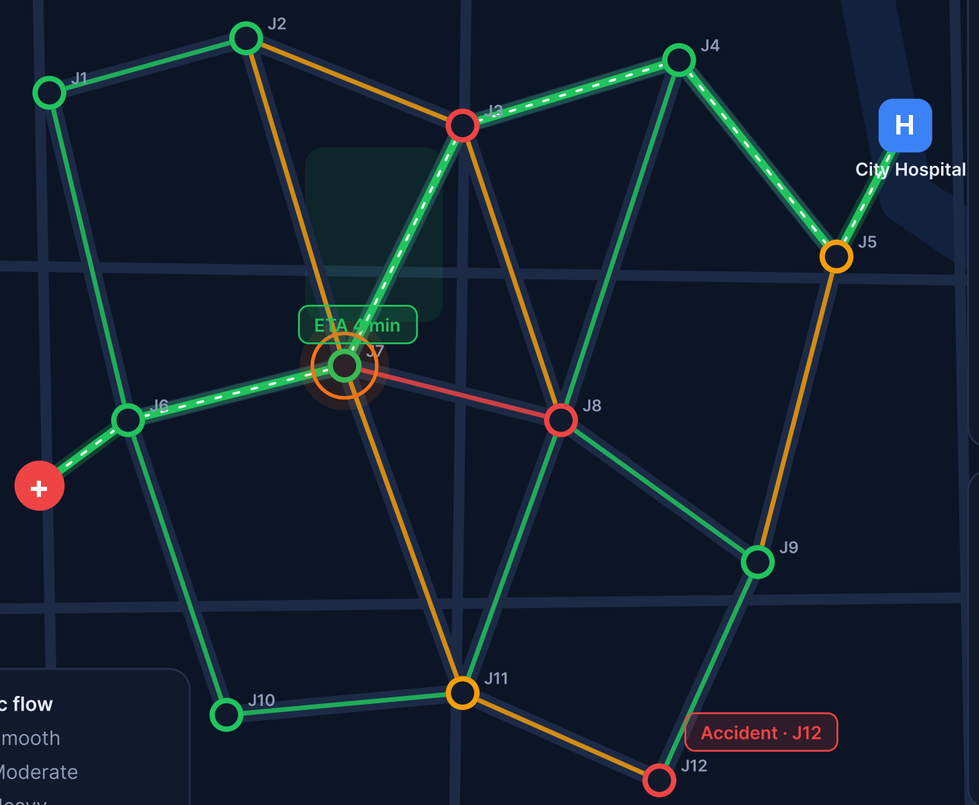
Active Incidents 1

- Accident detected — Junction J12 · 2 min ago — Verify

System Performance Demo data · last hr

Avg. waiting time	28 s ↓ 42%
Avg. travel time	6.3 min ↓ 35%
Throughput	3,420 veh/hr ↑ 18%
Prediction accuracy	92%
Incident detection	87%

Traffic | Signals | Cameras | Incidents | Emergency



Traffic flow
- Smooth
- Moderate
- Heavy
- Emergency route

System Mode

Normal | Emergency | Incident | Manual

✓ No conflicting signals · ✓ Min green kept · ✓ Pedestrian safe · ✓ Fallback ready

AI Decisions (live)

10:24:12 Extended green at J6 for Ambulance #A1

10:23:58 Predicted congestion at J12 (accident)

10:23:40 Rerouted 12% traffic to alternate roads

Selected Junction · J7

LIVE

CAM-J7-02 · YOLO detections

Current status	Green (28s left)
Next phase	Yellow (5s) → Red (42s)
Traffic density	High · 78 veh/min
Queue length	24 vehicles
Average speed	18 km/h
AI recommendation	Extend green +10s

Approve | Override

Signal Timing (dynamic)

North–South	42s
East–West	28s
Left turn	15s
Pedestrian	12s

AI Prediction · next 10 min

Congestion ~4m

PK — Priya Kulkarni — Traffic Operator · Sign out

Junctions

24 total

ID	Density	Cycle	Mode
J1	Low	72 s	Auto
J2	Medium	88 s	Auto
J3	High	96 s	Auto
J4	Low	70 s	Auto
J5	Medium	84 s	Auto
J6	Low	74 s	Auto
J7	High	97 s	Auto
J8	High	92 s	Manual
J9	Low	68 s	Auto
J10	Low	66 s	Auto

Timing Editor · Junction J7

Adaptive (network-level)

North–South green **42 s**

East–West green **28 s**

Left-turn phase **15 s**

Pedestrian crossing **12 s**

Cycle length (incl. all-red / yellow) **97 s (max 120 s)**

Run what-if simulation | Apply to junction | Reset

Safety Validation

4 / 4 passed

- Minimum green time (≥ 10 s) — ✓ Pass
- Pedestrian clearance (≥ 12 s) — ✓ Pass
- No conflicting phases — ✓ Pass
- Max cycle length (≤ 120 s) — ✓ Pass

AI proposals that fail any check are rejected and fixed-time fallback is used.

What-if · Digital Twin

SUMO · demo values

Avg. waiting time

Fixed-time baseline 52 s

UATMMS proposal **28 s**

Max queue length

Fixed-time baseline 41 veh

UATMMS proposal **24 veh**

Throughput

Fixed-time baseline 2900 veh/hr

UATMMS proposal **3420 veh/hr**

Neighbour impact (J6, J8) **+2% delay · bounded**

Emergency corridor **Compatible**

Approve & auto-execute

UATMMS
Traffic Intelligence

POLICE CONSOLE

- Command View
- Incident Queue

SP
Insp. S. Patil
Traffic Police · Sign out

Police Command View

Verify incidents · dispatch patrols · audited manual override

Search junction, vehicle...

System Online · AI Operational

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Patrol Units

4 units

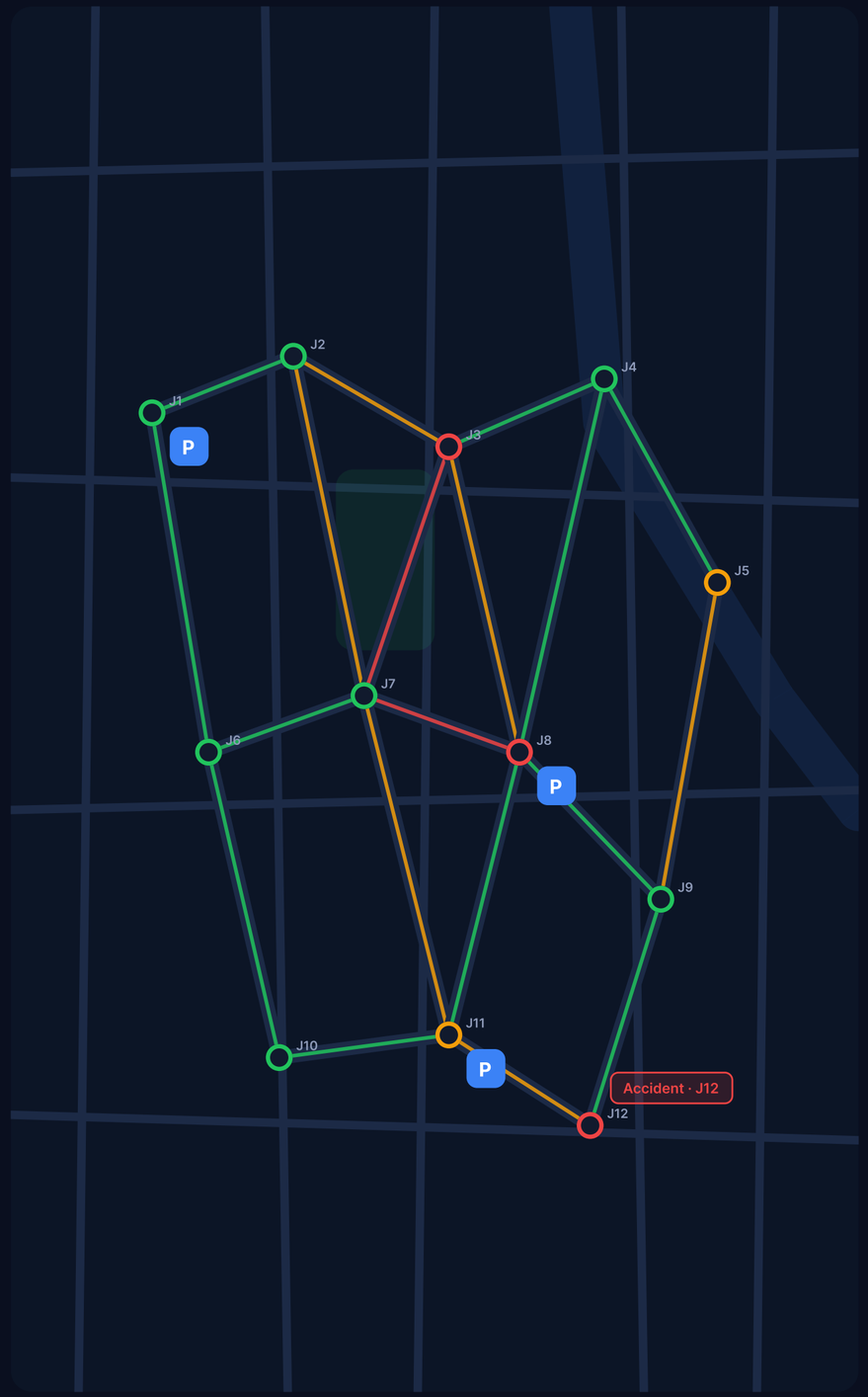
P-12 — Dispatched
En route to J12 · 2 min

P-07 — On scene
At J8 · stalled vehicle

P-15 — Escort
Escorting Ambulance #A1

P-03 — Available
Zone A · standby

+ Dispatch patrol



Verification Queue

2 pending

Accident · J12 — Critical
Vision AI 94% · 2 min ago
Verify | Dismiss

Abnormal slowdown · J3 — Medium
Anomaly model 76% · 18 min ago
Verify | Dismiss

Manual Override

Logged & audited

Junction

J12 · Gandhi Chowk — Change

- ○ Hold all red (emergency stop)
- ○ Flash yellow (signal fault)
- ● Manual phase control

Overrides pass through the Safety Engine — conflicting phases are always blocked.

Apply override (30 min max)

Incident Queue

All incidents detected by AI or reported by citizens

Incidents

Today

ID	Type	Location	Source	Severity	Status
#1042	**Accident**	J12	Vision AI · 94%	Critical	Pending
#1041	**Stalled vehicle**	J8	Vision AI · 81%	Medium	On scene
#1040	**Abnormal slowdown**	J3	Anomaly · 76%	Medium	Diverted
#1039	**Waterlogging**	J10	Citizen report	Low	Resolved
#1038	**Signal fault**	J5	Controller alarm	High	Resolved
#1037	**Road blockage**	J2	Police patrol	Low	Resolved

Incident #1042 · Accident at J12



Detected	10:22 AM · 2 min ago
Confidence	94% (ML + rules)
Lanes blocked	2 of 4
Diversion	Active · via J9 → J8 → J11
Nearest patrol	P-12 · ETA 2 min

Dispatch patrol

Call ambulance

Notify operator

Close incident

UATMMS

Smarter signals.
Safer roads.
Faster responses.

AI-powered, network-level traffic management with automatic green corridors for ambulances and fire trucks.

↓ 42%	↓ 38%	0
waiting time	emergency travel	safety violations

Demo targets from simulation — to be validated in pilot.



Welcome back

Sign in to your UATMMS console, or pick a demo role to explore.

Email / Employee ID

priya.kulkarni@kolhapur-traffic.gov.in

Password

•••••••••• Show

Sign in

or choose a demo role

Traffic Operator Web
Monitor & control the live network

System Admin Web
Users, policies, devices, analytics

Traffic Police Web
Verify incidents · manual override

Hospital / ER Web
Track ambulances · request dispatch

Driver / Commuter Mobile
Live traffic, alerts, reroutes

Ambulance / Fire Mobile
Request green corridor